Chapter-4

TU - 2023

(1) Briefly discuss the ~~affect~~ factors that affect crystal field splitting energy (Δ). (3)

Ans: Four factors:

(i) Valency or oxidation ~~of~~ state: বাড়লে Δ বাড়ে

(ii) Atomic no. of metal ion: বাড়লে Δ বাড়ে।

(iii) ~~Symmetry of complex~~ or geometry of complex: $\Delta_{sp} > \Delta_o > \Delta_t$

(iv) Nature of ligand: ligand যত বেশি strong Δ তত বেশি, Δ value of octahedral complex is double than (4) crystal field splitting of tetrahedral molecule. this is represented by the relation as $\Delta_{sp} > \Delta_o > \Delta_t$.

(iii) the no. of ligands and Geometry of complex: ligand যত বেশি হবে Δ বাড়বে sheet (দিলে বাড়বে) ~~বাড়বে~~

(2) Which complexes of the following pairs has a larger Δ value? Explain your answer: (6)

2017
2015 (i) $[Cr(NH_3)_6]^{3-}$ and $[CrCl_6]^{3-}$

Ans: According to the spectrochemical series NH_3 is stronger ligand than Cl^- and stronger ligand gives larger Δ value.

∴ ~~$Cr[NH_3)[Cr(NH_3)_6]^{3-}$~~ has larger Δ value.

(ii) $[Co(H_2O)_6]^{3+}$ and ~~Cr~~ $[Rh(H_2O)_6]^{3+}$

Δ value for corresponding complex of metals ions in the same group increases as the atomic no. of metal ion increases. Atomic no of Rh(45) is greater than Co(27).

So, $[Rh(H_2O)_6]^{3+}$ has large Δ value.

(iii) $[Cr(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{2+}$

In this case two complex has the same ligand but oxidation no of central metal atom ~~Co~~ in $[Cr(H_2O)_6]^{3+}$ is +3 and in $[Cr(H_2O)_6]^{2+}$ is +2.

The largest is the oxidation state greater is the Δ value.

So, $[Cr(H_2O)_6]^{3+}$ has largest Δ value.

(3) Explain the formation of the following on the basis of CFT; (5)

(i) paramagnetic $[FeF_6]^{3-}$

(ii) diamagnetic $[Ni(CN)_4]^{2-}$

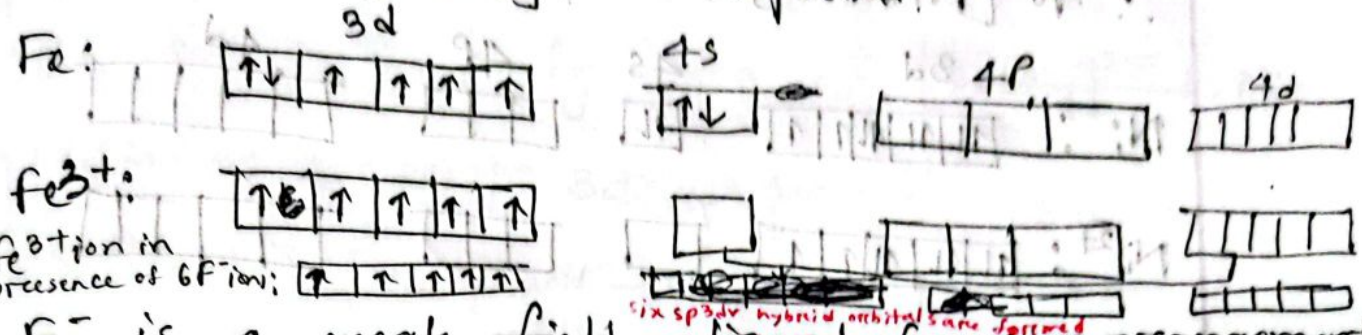
(iii) tetrahedral $[NiCl_4]^{2-}$

f-2021

(i) the electronic configuration of ~~Fe~~ Fe is:

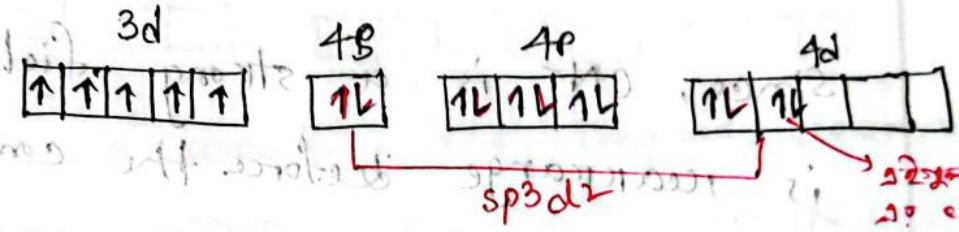
Fe(26): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ ~~3d^6~~ $4s^2\ 3d^6$

Now, for $[FeF_6]^{3-}$ compound:



F^- is a weak field ligand. So, rearrangement does not occur for Fe^{3+} in $[FeF_6]^{3-}$.

$\therefore$ $[FeF_6]^{3-}$:

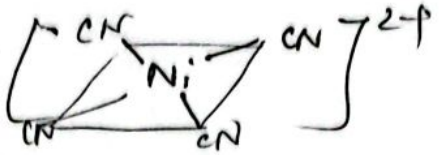


The hybridization of Fe in $[FeF_6]^{3-}$ is sp^3d^2 and the compound is octahedral. Since, there is unpaired electron present, the compound is paramagnetic in nature.

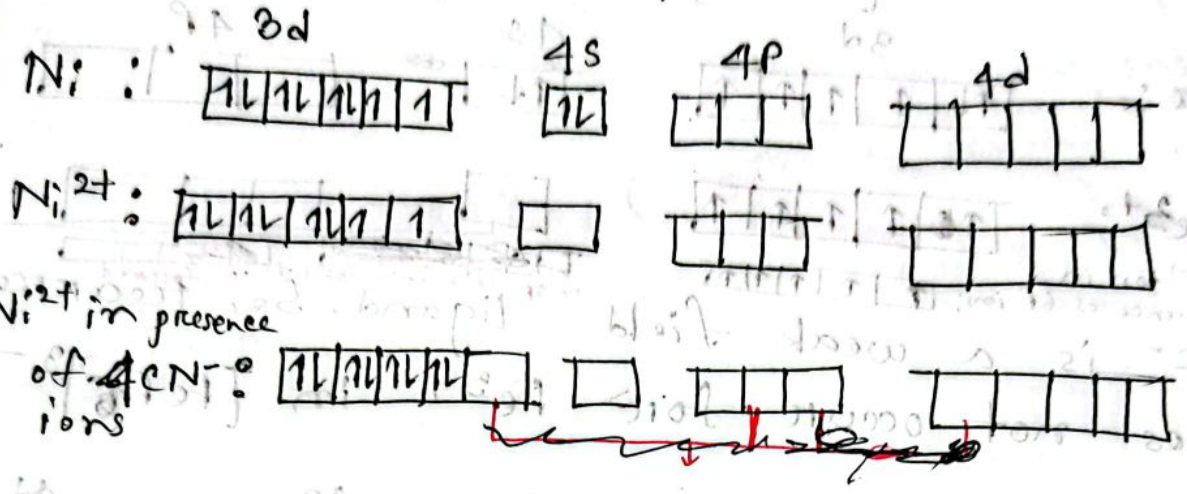
(ii) The electronic configuration of Ni is

Ni(28): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$

P.T.O.



∴ for $[Ni(CN)_4]^{2-}$ complex:



Since, CN^- is a strong field ligand, Ni^{2+} is rearrange before the complex ion formation.

∴ $[Ni(CN)_4]^{2-}$:

The hybridization of Ni in $[Ni(CN)_4]^{2-}$ is dsp² and the compound is square planar. Since, there is no unpaired electron, the compound is diamagnetic.

(iii) Tetrahedral $[NiCl_4]^{2-}$

The electronic configuration of ~~$[NiCl_4]^{2-}$~~ ~~K, Ni;~~

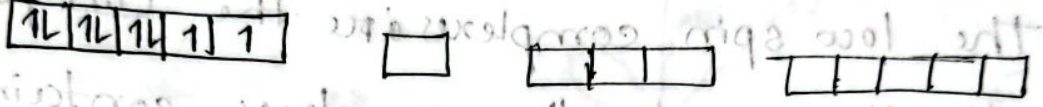
$Ni(28) \rightarrow 1s^2\ 2s^2 2p^6\ 3s^2 3p^6\ 4s^2 3d^8$

for $[NiCl_4]^{2-}$ compound –

Ni : 3d | 4s | 4p | 4d



Ni^{2+} :

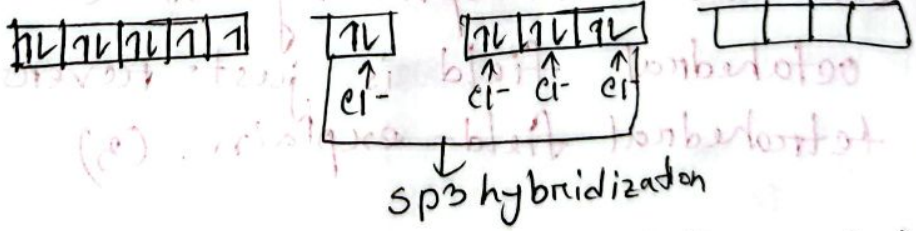


Since, Cl^- is a weak field ligand, rearrangement does not occur for Ni^{2+} in $[NiCl_4]^{2-}$

∴ Ni^{2+} in presence of four Cl^- ions



∴ $[NiCl_4]^{2-}$:



∴ The hybridization of Ni in $[NiCl_4]^{2-}$ is sp^3 and ~~it~~ has two unpaired electron. So, ~~it~~ $[NiCl_4]^{2-}$ has tetrahedral geometry.

N.B. [sp³ হলে ~~এটা~~ unpaired electron না থাকলে square planner হবে।]

(4) What are the high spin and low spin complexes? (2)

Ans: The high spin complex is the compound with the greatest number of unpaired electrons. Weak filed ligands lead to high spin complexes.

Example: $[FeF_6]^{3-}$

The loco spin complexes are the compound with the coordination complexes containing paired electrons at loco energy levels. Strong field ligand leads to the loco spin complex.

Example: $[Fe(CN)_6]^{3-}$

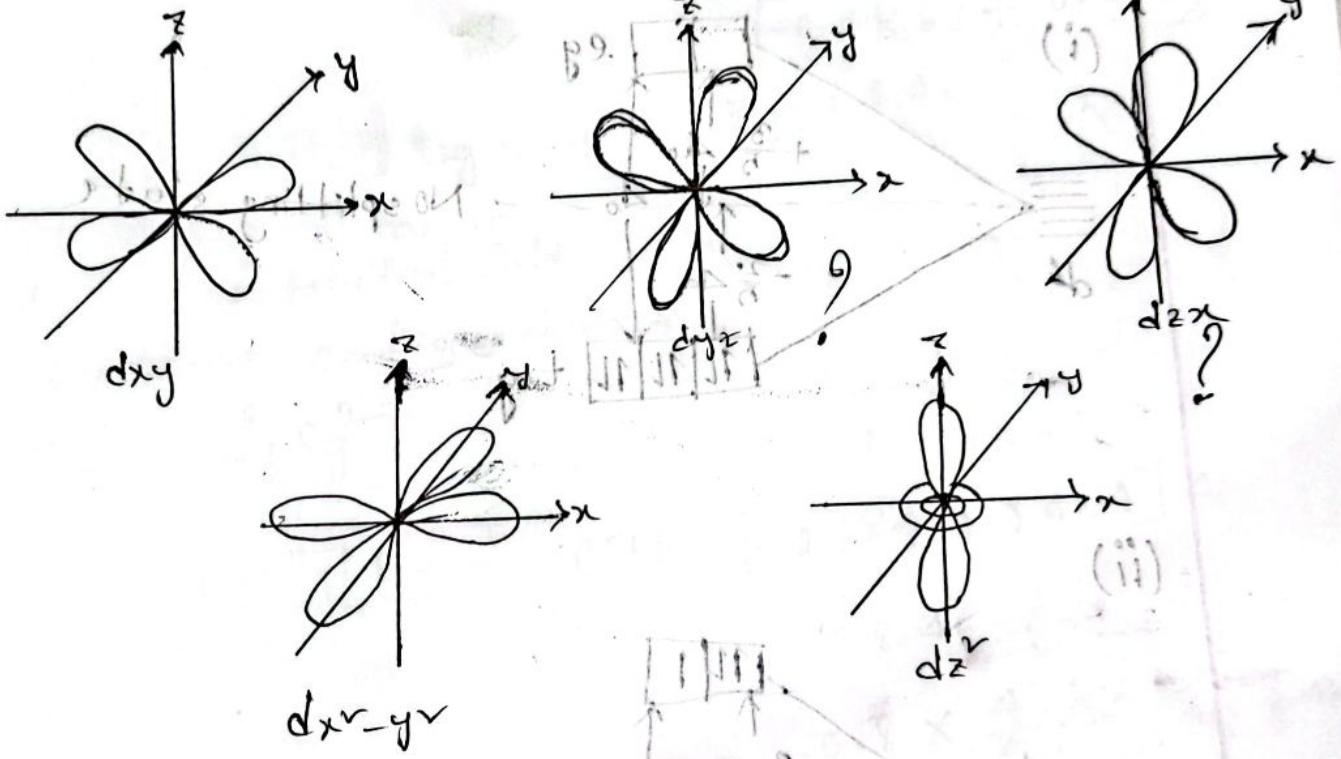
(5) Mode of splitting of d-orbitals in an octahedral field is just reverse of that in tetrahedral field - explain. (3)

Ans: Slide.

Back up Tutorial:

(1) Draco the shapes of the various d orbitals and explain why they split into two groups - t_{2g} and e_g in an octahedral field? (4)

Ans:



বাকিটা slide থেকে পড়তে হবে।

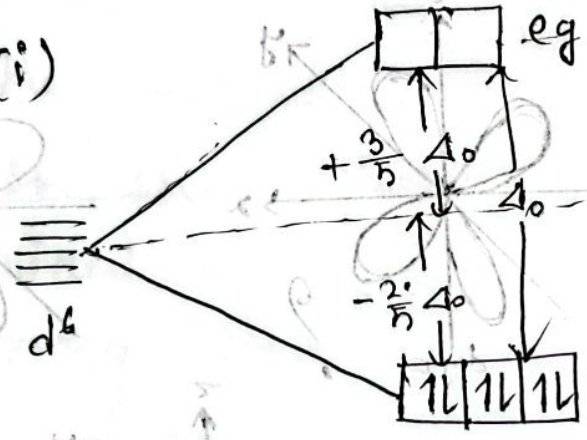
(2) Draco energy level diagram and indicate the occupancy of the orbitals in the following complexes. (6)

(i) d^6 octahedral low spin.

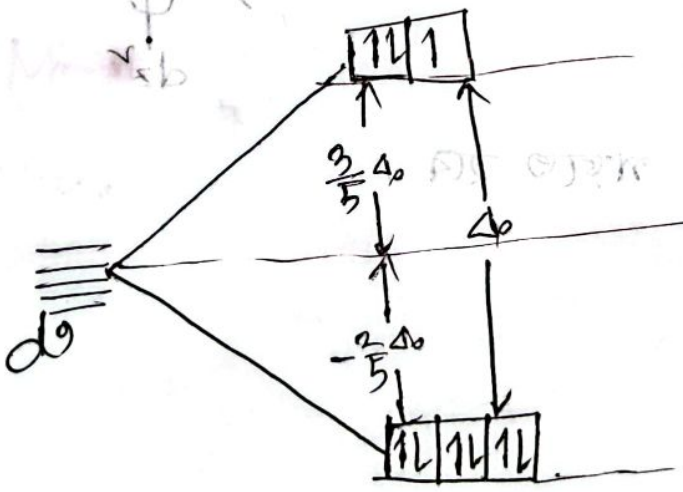
(ii) d^9 octahedral with tetragonal elongation.

Ans:

(i)



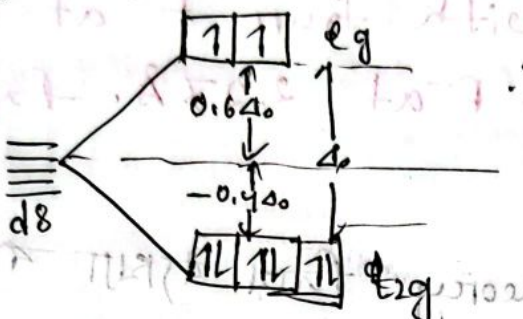
(ii)



2017

(3) Calculate the crystal field stabilization energy for d^8 ion in octahedral and tetrahedral complexes. Which one is more stable? — Explain (6)

Ans:



$\therefore CFSE = (-0.4 \times n t_{2g} + 0.6 \times n e_g)\Delta_o$

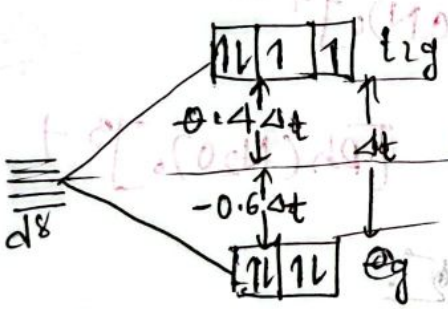
$= [-0.4 \times 6 + 0.6 \times 2]\Delta_o$

$= -1.2\Delta_o$

fig: d^8 ion in octahedral field

$\therefore$ Electronic configuration:

$d^8 \longrightarrow t_{2g}^6 e_g^2$



$\therefore CFSE = [-0.6 \times 4 + 0.4 \times 4]\Delta_t$

$= -0.8\Delta_t$

$= -0.8 \times \frac{4}{9}\Delta_o \quad [\Delta_t = \frac{4}{9}\Delta_o]$

$= -0.35\Delta_o$

fig: d^8 ion in tetrahedral field

$\therefore$ Electronic configuration:

$d^8 \longrightarrow e_g^4 t_{2g}^4$

In octahedral field, 6 ligands are attached to the d^8 ion and in tetrahedral field 4 ligands are attached to the d^8 ion. The more ligands are attached to the central metal ion, the complex

become more stable. So, d^8 in octahedral field is more stable. Again, CFSE value of d^8 ion in octahedral complex is greater than that of tetrahedral complex. Greater the CFSE value more it will be stable. So, d^8 ion in octahedral complex is more stable.

(4) In the crystal structure of CuF_2, Cu^{2+} is six coordinate with four F^- at a distance of 1.93Å and two F^- at 2.27 Å. Explain why? (4)

Ans: Jahn Teller theory

F-2021, (F-2018-5(a))

3. (a) Which complexes of the following pairs have a larger Δ_o value? Explain your answer.

(i) $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$

(ii) $[Co(NH_3)_6]^{3+}$ and $[Rh(H_2O)_6]^{3+}$ (4)

Ans: Physical khata

(d) What is crystal field stabilization energy? Explain the splitting of d-orbitals under the influence of an octahedral field. (3)

2017

Ans: Sheet.